CROSS-CULTURAL APPROACHES TO CONSCIOUSNESS

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CROSS-CULTURAL APPROACHES TO CONSCIOUSNESS

MIND, NATURE AND ULTIMATE REALITY

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CHAPTER FOUR

Abhinavagupta's *Svātantryavāda*: Mental Causality, Emergentism, and Intuitionist Mathematics

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1 INTRODUCTION

How do we get a concept of mind, indeed a mind that is not simply mere epiphenomenon but that can effectively cause events within our current paradigm of materialism—that is, within a paradigm which reduces causal relations to the interplay of physical objects, that is, neurons and molecules? The notion of mental causality, the idea that the mind can influence the material substrate is fraught, problematic for a materialist paradigm. On the one hand, much of our science follows an understanding that only matter can affect matter; mental constructions, whatever they are, cannot affect matter. For this scientific perspective, to suggest otherwise is to slide into the hocus-pocus of magic. On the other hand, the idea that mind does not affect the physical world—at a minimum inducing change in one's own physical body, as when a loving thought slows the heart rate—but even more broadly, that the mind does not impact the material world—this seems to lead to a view that goes against common sense. If what we think has no effect on the world or our bodies, that our consciousness, our minds are simply irrelevant, as Daniel Dennett famously argued, mere epiphenomenon (Dennett 1991), then in a sense we wind up erasing the impact of centuries of human culture, including even the human thought that has generated the very science we use to address the question. In other words, much is at stake, since a notion of mental causality is tied to the idea of free will, as we see with John Searle famously lifting his arm as an instance of both (Searle 2006). Here, the crux of the question is *how* might what we think of as thought manage to interact with and affect materiality? To get around the problem of mechanism, a number of contemporary materialist philosophical renditions rely on a concept of emergentism as a way of supporting a link between the mental and matter so as to render mental causality efficacious. To put this, for the sake of brevity, much too simplistically, if mind *emerges* out of matter, then its seeming difference from matter is not an ontological barrier to downward causality, where mind can interact with materiality.

Of course, mental causation as a concept relates to a variety of formulations addressing the nature of the mind-body relation, for instance, that certain mental patterns can generate physiological effects, such as heart rate or brain chemistry release of dopamine, or more radically, that the mental mantra practice of a yogi can prevent contact with fire from burning her skin, or that a yogi like Sai Baba can generate a mūrti, a statue of Kṛṣṇa through thought alone, and so on. For the purposes of this paper I will focus on only one particular philosophical consequence of mental causation, the aspect of free will. Even as the idea of mental causality encompasses a number of different components, addressing the component of free will as one particular aspect of mental causality highlights what I suggest we can see as an intrinsic link between a subjective perspective and mental causality.

With this, I offer a different avenue than emergentism for linking mind to materiality and thus enabling a capacity for the mental to affect the physical. I draw on Abhinavagupta's eleventh-century articulation of *Svātantryavāda*, the "Free-will Doctrine" to point to a model for mental causation that does not rely on emergentism; instead Abhinavagupta's model proposes a modal argument based on a foundational subjectivity as a means of positing mental causation. I suggest that Abhinavagupta offers a model where mind itself, consciousness as a first-person subjectivity is a foundational category that generates the materiality of the world we encounter. This operates as a type of cosmopsychism, which unfolds out of itself the materiality that makes up the physical world.

Certainly, proposing a foundational subjectivity is fraught; a perspective of subjectivity is frequently criticized as being insufficiently scientific, with the implicit equation of science with a third-person, objective perspective. This criticism, I suggest, is unfounded; to support this, I draw from the work of physicist Nicolas Gisin and before him, L. E. J. Brouwer's articulations of Intuitionist Mathematics (Brouwer 1981). What is interesting about this alternative path to mental causation, based within a concept of free will, I suggest, is that Gisin and Brouwer argue for its basis within a rigorous mathematical model that does not depend on emergentism, or for that matter, on panpsychism. I suggest that we find a similar framework to Gisin's use of Intutionist Mathematics, and more especially Brouwer's entailment of subjectivity with his idea of the "Creating Subject," in Abhinavagupta's *Svātantryavāda*, and that Abhinavagupta's model, intuitively derived, follows a similar logic. Abhinavagupta is not a mathematician (and nor am I), and his own articulation of his model of consciousness involves no mathematics; however, his model shares structural similarities to these mathematical articulations, specifically: The inclusion of free will, a non-deterministic system, a necessary inclusion of the idea of time. What Abhinavagupta's *Svātantryavāda*, Free-Will Doctrine adds to these mathematical derivations is to make explicit the emphasis on subjectivity and a first-person perspective, an element not found in Gisin's articulation, which is, however, incipiently articulated in Brouwer. With this, I suggest that we ought not to discount subjectivity as a model simply because of a bias that sees objectivity, a third-person perspective as more scientific or logically precise. This also suggests that Abhinavagupta's medieval adoption of a subjectively oriented philosophical model can add to the conversation on causality[1] and that we might find productive results in unpacking this alternative path toward accounting for mental causation.[2]

2 PANPSYCHISM, COSMOPSYCHISM, EMERGENTISM

While panpsychism was developed as a way around the hard problem, recently a variety of thinkers have adopted forms of panpsychism as a way to entail mental causation while still avoiding the philosophical problems accompanying emergentism.[3] As Godehard Brüntrup and Ludwig Jaskolla point out, panpsychism offers "a world view which is capable of remaining within a broadly conceived naturalist framework whilst accounting for the emergence of a nonreductively conceived mental reality" (Brüntrup and Jaskolla 2016: 3). Panpsychism, since it entails an already embedded notion of mind within all matter, seems to stave off the problem of needing an emergentist doctrine, or at least a strong emergentism, to account for mentality and mental causation. Nevertheless, various models that embrace panpsychism still invoke a form of emergentism, particularly as a way around the combination problem.[4] Giulio Tononi and Christof Koch's Integrated Information Theory 3.0 (IIT 3.0) is an illustrative case, asserting panpsychism, while relying on a form of emergentism to distinguish between objects that display consciousness (*Phi* in their terminology) and those that do not. Tononi, for instance, proposes that we might find consciousness even in a simple mechanical device, a photodiode. Underwritten by a foundational panpsychism, actual, measurable consciousness (*Phi*) emerges only in instances where information as self-reflexive activity is demonstrated. The operating principle in this model is that if enough information as bits is accumulated and nested within integrated circuits of recursive feedback loops, meaning will emerge. A jump to consciousness will occur.

Describing the process in an imaginary dialogue between Galileo and Leibniz, Tononi writes, "which mechanisms would one have to add in order to augment the lowly photodiode and make it like Galileo's brain?" (Tononi 2012: 199). In order to get, as Tononi puts it, "meaning out of mechanism," the Baron tells Galileo,

> By piling one mechanism on top of another, that's how it's done! … One mechanism alone—things are this way and not another way—generates hardly any information: one nondescript bit. But if you pile them up, so that one builds upon another, not only side-by-side, but each standing on the shoulders of the others, as it were, they generate far more information, bits multiply.
>
> (Tononi 2012: 203)

With enough nested circuitry one can make the jump from the mere mechanism of a manufactured mechanical device, a photodiode, to the wonder of consciousness. That is, Tononi's method entails a kind of emergentism: "if you pile them up" some new shift occurs; consciousness emerges. Tononi has made it clear that he subscribes to a model of panpsychism; yet he incorporates emergentism to explain why it is that even though consciousness is all-pervasive, objects like a cellphone do not exhibit the self-recursivity required for what we think of as consciousness (Oizumi, Albantakis, and Tononi 2014) and (Tononi n.d.).

Abhinavagupta, on the other hand, does not subscribe to an emergentist model.[5] This is where his panentheism—which again might be considered akin to what thinkers like Itay Shani, Joachim Keppler, and Philip Goff designate as cosmopsychism, a kind of overarching single principle that spreads itself out into what makes up the world—diverges from the type of micro-panpsychism Koch and Tononi embrace.[6] In this respect we might understand his panentheism as a form of cosmic consciousness. Shani argues that it is possible to formulate a cosmopsychism that avoids the combination problem entailed in micro-panpsychism and can also preempt its mirroring obverse, the decombination problem (Shani and Keppler 2018). While it may be possible to argue that

Abhinavagupta's model evades the combination and decombination problems by virtue of its adoption of a singular cosmic consciousness, I focus here explicitly on the links his model proposes between free will and subjectivity as the mechanism for mental causality.

In other words, embedded within Abhinavagupta's conception of cosmic consciousness is a separate avenue for arguing for mental causation: his prioritization of subjectivity, with a concommitant freedom, over a third-person perspective.[7] Thus, he tells us elsewhere, "this in fact is freedom, which is the capacity to do that which is exceedingly difficult. And that difficulty is to be actually placed on the level of *Māyā*."[8] The idea of freedom, which is intrinsic to subjectivity as modality, is what allows a singular consciousness to individuate into a multiplicity of different conscious entities each with their unique capacities to exercise divergent expressions of free will.[9] Thus, he tells us: "When those [beings] belonging to *Māyā*—even down to a worm, an insect—when they do their own deeds, that which is to be done first stirs in the heart."[10] That is, even the tiniest insect has an agency and a consciousness capable of choosing, a subjectivity rising in the heart. Specifically, by subjectivity as modality I mean that Abhinavagupta suggests a phenomenology of the first-person perspective at its base entails an essential freedom that serves as foundational for subsequent truth claims.[11] In what follows, we look comparatively at the conditions for this freedom in Abhinavagupta's model in relation to Gisin's adoption and Brouwer's articulations of Intuitionist Mathematics. Beginning with Gisin's articulation of the problem of real numbers, as the mathematical argument for this alternative model, we then address specifically three areas of convergence Abhinavagupta shares with Intuitionist Mathematics: Indeterminism, its links to freedom, and the inclusion of time. After this we examine Abhinavagupta's postulation of subjectivity.

3 REAL NUMBERS

Gisin points to the origin of Intuitionist Mathematics in an early twentieth-century debate between mathematicians L. E. J. Brouwer and David Hilbert on how we might count numbers. Hilbert insisted that we understand every real number with its infinite series of digits as a completed individual object (Gisin 2020: 114). Brouwer, arguing for an Intuitionist Mathematics, took the view that each point on a number line should be represented as a never-ending process that develops in time (Gisin 2020: 114). As Gisin relates, the question has to do with whether mathematics exists within time or whether it operates within an idealized Platonic realm, and this mathematical debate mirrored another debate also focusing on time, and happening around the same time in Paris (Gisin 2020: 114). In this related debate, Albert Einstein debated the French philosopher Henri Bergson (Gisin 2020: 114). Bergson critiqued Einstein's conception of time prompting Einstein's famous assertion "Il n'y a donc pas un temps des philosophes" (The philosopher's idea of time does not exist) (Canales 2015: 5). Even as Einstein's argument against Bergson eventually eclipsed Bergson's position, at the time Bergson's critique that Einstein's idea of relativity and its assertion of time pertained not to physics, but to epistemology, hit a nerve.[12] So much so that it swayed the Nobel Prize committee to give Einstein the Nobel prize for his work on the photo-electric effect, not for what made him famous at the time, his celebrated formulation of relativity (Canales 2015: 3–4).

Perhaps aware of this other debate, Brouwer's intuitionist mathematics focused on understanding mathematics as embedded within a schema of unfolding time. Precipitated by a logic problematizing an infinite series of digits as ultimately a product of random selection, Brouwer formulated a mathematics tied to a materially embedded reality, operative expressly within the parameters of time and the free choices made by the mathematician engaging a mathematical problem (van Atten 2018). Gisin, whose own work spans classical and quantum physics, draws on Brouwer particularly in relation to the problem of time, infinity, and information for a contemporary model. Gisin points to the problem of a mathematically projected infinity in relation to a finite volume. He says, "no finite volume of space can contain an infinite amount of information. This is a well accepted result that follows from the holographic principle, known as the Bekenstein bound" (Gisin 2019: 6). For Gisin, a mathematical stipulation of real numbers as infinite carries its own burdens, in particular, an unwieldy infinitude at the heart of physics. He proposes as remedy an alternative mathematical physics based on Brouwer's Intuitionist Mathematics. This alternate mathematical system recognizes the disjunct, the actual practical impossibility of a mathematics that operates on an infinite amount of information contained between two numbers on a number line. In a nutshell: The impossibility of containing an infinite amount of information within a finite space.

Gisin derives his formulation from Brouwer's Intuitionist Mathematics. As Brouwer outlines the basic principles, the problem is that the history of mathematics in the nineteenth and twentieth centuries introduced a formalism divorced from an actual experiential engagement with mathematics. Intuitionist mathematics proposes recognizing an intuitive understanding of mathematics grounded in experience and developed intuitively, separate from the formulation of a language to describe mathematics, instead understanding mathematics as "an essentially languageless activity of the mind having its origin in the perception of a move of time" (Brouwer 1981: 1–4). As such, Brouwer's formulation entails that certain logical conditions that we take for granted, such as the law of the excluded middle, are simply artefacts of a historical construction and only applicable in certain situations. (Brouwer 1981: 5–8). To return to Gisin's formulation of the problem of an assumed mathematical infinity, Gisin tells us:

> Today, everyone knows that they hold gigabytes of information in their pockets … Furthermore, everyone knows also that each stored bit requires some space. Not much, possibly soon only a few cubic nanometre (10^{-18} mm^3) … Consequently, assuming that information has always to be encoded in some physical stuff, a finite volume of space cannot contain more than a finite amount of information.
>
> (Gisin 2019: 7)

This is his basic premise, that information occupies space. Information encoded within a material substrate is limited, even as our current predominant mathematics offers a system predicated on infinite information. Further:

> Consider a small volume, a cubic centimetre let's say, containing a marble ball. This small volume can contain but a finite amount of information. Hence, the centre of mass of this marble ball can't be a real number … since real numbers contain—with probability one—an infinite amount of information … the assumption that a finite volume of space can't contain more but a finite amount of information implies that the centre of mass of any object cannot be identified with mathematical real numbers. Real numbers are useful tools, but are only tools. They do not represent physical reality.
>
> (Gisin 2019: 7)

As he outlines it, his formulation of Intuitionist Mathematics works much the same as classical mathematics; its advantage lies in its acknowledgment of this basic incompatibility—between a Platonic mathematics of infinity and our current real-world awareness that a finite space can only contain a finite amount of information. With this he proposes a mathematics and physics that operates similarly to classical mathematics and physics, but which addresses the problem of infinite information in a finite space:

> In a nutshell, this alternative theory keeps the same dynamical equations as classical mechanics, but all parameters, including the initial conditions are given by numbers containing only a finite amount of information … I argue that this alternative classical mechanics is more natural because it doesn't assume the existence of inaccessible information.
>
> (Gisin 2019: 2)

Even as Gisin's alternative mathematical theory functions similarly to classical mathematics in solving physics and mathematical problems, it nevertheless carries with it a number of philosophical implications. I will address the implications below: These include 1) a model that suggests indeterminism even within classical physics, not just within quantum physics where it is familiar; 2) this model is tightly linked to an assertion of free will, again, resonant with Abhinavagupta's philosophical model of Recognition (*Pratyabhijñā*), which has alternatively been called the Free-Will Doctrine, *Svātantryavāda*; 3) directly following from indeterminism, the reincorporation of time within physics.

4 INDETERMINISM LINKED TO FREEDOM

Many of the philosophical ramifications of Gisin's alternative mathematical theory follow directly from postulating a primary indeterminism. Gisin derives this from his initial premise that a finite amount of space cannot contain an infinite amount of information. The finitude of physical space links back to Brouwer's intuitionist mathematics and Brouwer's assertion that the sequences of numbers generated on a continuum are not infinite, but instead generated as random numbers (Gisin 2020: 115). As Gisin relates, understanding the fundamentally uncomputable nature of real numbers means that we should understand them instead as random numbers; the term "real numbers," he tells us, is a misnomer:

> In other words, almost all real numbers are random in the sense that their sequences of digits (or bits) are random … And these random numbers (a better name for real numbers) should be at the basis of scientific determinism? Come on, that's just not serious! … it is also a great source of confusion in the philosophy of science.
>
> (Gisin 2016: 5)

That is, the very basis of our mathematical system leads us astray, inclining us to miss the fundamentally random nature of real numbers. That they are random means that they cannot be used to predict determinate outcomes; the random quality of these numbers will yield only imprecise, indeterminate mathematical results.The consequence is that scientific determinism has no solid foundation, no real leg to stand on. This is where his alternative mathematics adheres more closely to the actual situation of things. What it means is that at base, even for classical physics—not just for quantum physics—the notion of a scientific determinism does not follow our material situation. Gisin is quick to point out that these parameters, determinism vs indeterminism, are elements built into the particular model of mathematics we use:

> One may object that intuitionism doesn't derive indeterminism, but assumes it from the start. That is correct. Likewise, classical mathematics assumes actual infinite information from the start. These two views cannot be distinguished empirically. One can always claim that instead of God playing dice every time a random outcome happens, God played all the dice at once before the Big Bang and encoded all results in the Universe's initial condition. (Gisin 2020: 115)

Pointing out that Einstein's oft-quoted rejection of quantum physics' structural indeterminacy is simply kicking the can down the road, back chronologically to the Big Bang, Gisin identifies the postulation of infinite information as an assumption within classical mathematics. Thus, this alternative, Intuitionist Mathematics instead takes as its premise indeterminism; God left some of the dice-play for after the Big Bang. Gisin justifies choosing indeterminism as a premise by pointing to the equally unsubstantiated and problematic presumption of our usual mathematics, that it assumes infinite information from the start. And as we saw, the problem with an infinity of information is that it is not supported by what we actually know about information, that a finite space can only contain a finite amount of information.[13] This also entails as a foundation for Gisin's argument that there is a real world and that physicists model their calculations on this real world, not on a mathematics that proposes infinite information in real numbers.[14] His alternative Intuitionist Mathematics is more closely aligned to the idea of the world as real and physically tangible, because, "it doesn't assume the existence of inaccessible information"(Gisin 2019: 2).

Abhinavagupta's model similarly, unlike an Advaita Vedānta ascription of the world as illusion, emphasizes that the objects we encounter here are real.[15] As K. C. Pandey pointed out in early scholarship on Abhinavagupta, Abhinavagupta's philosophy tends to work against our expectations of ontology; we tend to limit our choices to nondualism as idealism on the one hand, or else an actually existing world with a plurality of really existing different entities. Abhinavagupta subverted this false choice in a kind of have-your-cake and eat-it-too, what Pandey in a seemingly oxymoronic gesture labels a "Realist Idealism."[16] Elsewhere I argue that we might understand Abhinavagupta's nondual Śaivism precisely as a reclamation of a very real materiality (Biernacki 2022). In his eleventh-century *Īśvarapratyabhijñāvivṛtivimarśinī*, he reminds us, "the object as something real is not to be denied."[17] The materiality of the world is elemental to his philosophy of agency; as he tells us, "when you separate off the effects [of creation, the body, etc.] then the state of Doership does not occur."[18]

The idea of "Doership" (*kartṛtva*) as activity in the world references the power of the absolute as consciousness to connect to the materiality of the world; it is this element that differentiates Abhinavagupta's system from an idealist system like Advaita Vedānta which sees the world as insubstantial (*Māyā*). In practical terms, with the idea of "Doership" Abhinavagupta argues for an intrinsic link between consciousness and matter and precisely, the capacity of consciousness to generate material effects, that is, the capacity for mental causality.[19]

To return to Intuitionist Mathematics, even as it works empirically and is indistinguishable from classical mathematics on an empirical level, it carries profound philosophical ramifications. It radically alters how we think of the world. It means that our world is not predetermined, not just for quantum events, but even for a classical physics. Intuitionist mathematics tells us that real numbers are actually random and thus not capable of offering determinate outcomes. As Gisin puts it, "Here indeterministic is merely the negation of deterministic … given the present and the laws of nature, there is more than one possible future" (Gisin 2019: 3, n.2). This means as well that our models of physics and the other sciences, including for instance a neuroscience that proposes a capacity to determine outcomes mathematically, algorithmically, these algorithms will at base—that is, in principle—be incapable of fully determining an outcome. And not simply on an epistemological level. In fact, ontologically, our world is not predetermined.[20]

Gisin also explicitly links this indetermination to freedom. He tells us, "our acts of free-will are beginnings of chains of consequences. Hence, the future is open, determinism is wrong" (Gisin 2016: 2). Similarly, pointing to the link between indeterminacy and free will, Gisin argues, "The existence of genuine free-will, i.e. the possibility to choose among several possible futures, naturally implies that the world is not entirely deterministic" (Gisin 2016: 3). Ultimately, Gisin supports his argument for free will with his mathematical demonstration that real numbers should be understood as not real, but random numbers. As he puts it:

> the tension between free-will and creative time on one side and scientific determinism on the other side dissolves once one realizes that the so-called real numbers are not really real … the real numbers that theories use as inital conditions and parameters are not physically real. Hence, neither Newtonian, nor relativity, nor quantum physics are ultimately deterministic.
> (Gisin 2016: 2)

Further, like Abhinavagupta, he makes the idea of free will foundational, saying, "Free-Will comes first in logical order" (Gisin 2016: 2).

Abhinavagupta's cosmology encodes indeterminacy as well, though not through a temporal predictive model. As with Gisin, indeterminacy is foundationally linked to free will, however, Abhinavagupta frames indeterminacy as a corollary to subjectivity, "I-ness" (*ahantā*). Subjectivity is the basic reality; it entails an originary state of plenitude, a dense mass of consciousness (*cidghana, cidānandaghana*) as undefined and indeterminate, in contrast to "This-ness," the state of being an object.[21] However, Abhinavagupta understands indeterminacy entirely in psychological terms. Driven by an overarching process that leads us astray—*Māyā*—this operation of *Māyā* acts as a determinate form of thinking which limits our capacity to both know and to do things in the world. *Māyā* is the process of individuation, a determination, delimitation, and limitation of activity. Abhinavagupta quotes the Rudrayāmala text, saying, "*Māyā* is the name of the one who deludes, who is seated in the process of individuation, the limitation of activity," ("māyāvimohinīnāma kalāyāḥ kalanaṃ sthitam |" iti hi śrīrudrayāmalasāravākyam.) (Abhinavagupta 1985a: 3:285). An original *indeterminate* reality linked to subjectivity is lost when *Māyā* begins to operate. The mind distinguishes one thing as different from another (*vikalpa*), making determinate what was indeterminate, as a phenomenology of mental processes. Even as Abhinavagupta's model is primarily based on a mental phenomenology, still, for Abhinavagupta as well, this also means that events are not predetermined. And even though our karma plays a role, still there is no fate already set in motion, impervious to our free-will choices, where we just go through the motions.[22] The freedom (*svātantrya*) tied to subjectivity has genuine creative effects. Real creativity is indeed possible, even for limited beings, such as us humans, as Abhinavagupta makes clear: "'or by his own independence'—accepting the idea that there is creation which is not dependent upon the creation of the Lord."[23] Indeed, this is the case even for a lowly insect. As Abhinavagupta tells us, "When those beings belonging to *Māyā*—even down to an insect—when they do their own deeds, that which is to be done first stirs in the heart."[24]

The structural similarity Abhinavagupta shares with Gisin's model is thus: Where we find indeterminism, we find freedom, and the converse as well. Abhinavagupta, on the other hand, charts the link in terms of mental processes. In a colorful metaphor that emphasizes the mental projection involved, that is, the degree to which a sense of subjectivity is a habit of mind, Abhinavagupta likens indeterminate subjectivity as freedom to the black soot that covers a lamp. If it gets differentiated, parceled out into separate bits, or moved from the lamp to some other spot, the blackness of the soot makes a mess; it is no longer pure, no longer free, yet in its own place on the lamp, as an undifferentiated mass of blackness, it is pristine.[25]

To reiterate, Abhinavagupta's model is not a mathematical formulation, and not based on Gisin's understanding of the finitude of information. Rather, it is tied to the idea of subjectivity; again, in this case what he shares with Gisin's model of indeterminacy is the link between indeterminacy and freedom. Unlike

Gisin's use, Abhinavagupta's model is prescriptive, not a map that tells us how the world is; indeterminacy is a state of mind that enables a greater capacity for freedom. I suggest he probably derives his model from the meditation practices of Tantra, charting the mind phenomenologically as it moves from an initial indeterminate awareness to concrete conjectures as a mental process. Abhinavagupta (Abhinavagupta 1985a: 3:330) tells us:

> Free awareness, that is free, and awake. That is the name given to the form of the one who is aware. That which is called "I", in thinking "Who is that?" when with the body etc.—the form of that I becomes filled up with ideas like, "I am the one who is white;" "I am the one who is hungry;" and so on, then, the body and so on, alone become predominant, displacing the knower's true form and status as pure consciousness. So the object alone becomes predominant, not the form of the knower.
>
> But when he sets out to direct thought to the form of pure consciousness which is one unified mass of light, asking himself, "what Archetypes make up the body etc.?," then the Emptiness and the body etc., are no more as they are made into the true real form which is awareness, pierced through with the juice of consciousness. By practicing this, by properly entering into this awareness, he gains the special Energies which are the attributes of pure consciousness, and the miraculous yogic powers arise. However, even without practicing it, just by entering into it only for a moment, bliss rises up, along with the visible manifestation of other yogic experiences such as bliss, trembling, deep slumber, a feeling of pervasiveness, and whirling.[26] These occur one after the other and he attains liberation for his soul even while still alive in the physical body.[27]

Thus, we see here that Abhinavagupta links freedom to subjectivity specifically in terms of a phenomenology of the mind. This freedom also manifests as the *siddhis* of the yogi, the powers to manipulate material objects, which we find in the hagiographical tales of yogis inhabiting dead bodies, and changing matter.[28] When the sense of "I" becomes limited as "I am hungry" and so on, one takes on the status of being object, losing one's true form as undifferentiated subject. However, indeterminacy and determinate status are driven by subjective awareness and as such, can be influenced by mental processes and mental choices. Developing a particular habit of mind, entering into an awareness of consciousness as an undifferentiated mass of light, one is then able to acquire yogic abilities. Moreover, even without exerted practice, simply experiencing this awareness, Abhinavagupta tells us, leads to states linked to yoga, such as bliss and certain movements of the body such as trembling and so on. In this way, indeterminacy as a marker of a particular awareness of subjectivity is primarily a phenomenological process tied to freedom, and linked with yoga and tied to yogic abilities as its expressions.

5 TIME

To bring this back to cosmology, we also find time reincorporated back in, as a central feature, in both Gisin's and Abhinavagupta's models. For Abhinavagupta this is important since his cosmopsychism stands in contrast to earlier, alternative forms of nondualisms, such as Advaita Vedānta, which situate subjectivity within an idealized time-free or time-transcendent reality—much as we see with Gisin's characterization of classical mathematics as Platonized and outside of time. In his article "Time Really Passes, Science Can't Deny That," Gisin tells us "[n]on-determinism implies that time really exists and really passes: today there are facts that were not necessary yesterday, i.e. the future is open" (Gisin 2016: 2). He bases his argument on the argument outlined earlier: A finite amount of space cannot contain an infinite amount of information and that real numbers are essentially random numbers. As a consequence, he tells us the future is not predetermined. His insistence on this indeterminism, an open future itself entails free will. A consequent corollary following from these two is the reality of time. Gisin nuances his argument on time in this regard, allowing for two types of time, what he calls Einsteinian- or Parmenides-time, the type of time amenable to classical mathematics as "geometric-boring time" where no change occurs, and where using real numbers works fine, and Heraclitus-time where actual change occurs, and chance with its indeterminability is a factor (Gisin 2016: 6). With this, he gives a system accommodating both the idea of time as symmetrical and a pre-determined universe as Einsteinian-time and also the reality of time as a genuine unfolding of events allowing for chance and free will. For Gisin, chance and the possibility of actual newness, the open indeterminate future, Heraclitus-time, can and does occur, while at other times Einsteinian- or Parmenides-time gives us the predictable world, which is mostly, but not always what we experience. Abhinavagupta as well reincorporates an idea of time, in his case as a corollary of a first-person embodiment. Like Gisin, Abhinavagupta pushes back against the idea of timelessness by connecting time to free will, in his articulation expressed as intentionality. Intentionality, as an orientation toward one thing or another is the heart of a capacity to will. And this, he tells us, already embeds the fact of time. I give here a long passage which outlines his position:

> 'The intent of the will is the first instant of time' (*Śivadṛṣṭi* 1/7–1/8). Even the highest Reality, *Śiva*, the energy of bliss—which is that form where unfolding is just about to happen—that energy of bliss has priority, i.e., primacy, and prior existence with regard to the energy of will, [i.e., there *is* sequence here as well]. The word '*tuṭi*' is explained as a unit of time. Otherwise in the absence of sequence how could there be states of priority and posterity? However, with reference to those who are to be instructed and to those limited souls who are completely within [the grip of *Māyā*], there [teaching a lack of sequence] is appropriate.[29]

Here, Abhinavagupta argues against the idea that some absolute reality, or deity, might exist outside of time, beyond the reach of time. What Abhinavagupta suggests here does go against a dominant reading of Abhinavagupta's nondual Śaivism, where it is aligned to an idea more akin to Advaita Vedānta, with a space transcendent of time. What I suggest here is that if we read Abhinavagupta closely, we see that he realizes the philosophical problem embedded within a view that offers a transcendent space outside of time. As I suggest elsewhere (Biernacki 2022), for his nondualism to really incorporate the world, it must also incorporate time. And this is precisely why he brings the notion of time back in, explaining that the idea of a transcendent space outside of time is really just a skillful teaching tool for those who are caught in *Māyā*. At base, however, time occurs at the very heart of the highest deity because that deity, Śiva, has the power to will. The capacity to will, as intention, that is, as free will, is itself indicative of time. Those schools of thought that locate a transcendent reality outside of time, he tells us, are simply catering to students whose understanding is limited. This works fine as expedient teaching, acceptable for folks who are caught in the illusions of *Māyā*, but it is not the truth. He continues:

> You may protest that it says here in this verse that limited souls which are created have sequence; [since they exist within time], but the *Śiva* Archetype does not. In reply [Utpaladeva] says "because of the universality of the Subject." There also, in fact, the presence of sequence is proved. So [the text] says "that, in this way".[30]

That is, the Subject, who resides in the limited *Māyā*-bound individual is the same as the Subject which is *Śiva*. If the limited Subject can manifest sequence, surely the highest reality *Śiva* has the capacity also to contain sequence.[31] The essence of his argument depends on the idea that subjectivity is foundational and all-pervasive, in both the limited person and the highest absolute. Even if we can postulate some absolute reality or transcendent deity, outside of time, the very notion of a foundational subjectivity entails that it instantiates everywhere, all-pervasively.[32] This means that this subjectivity at base entails free will and that just as we in our limited bodies experience the unfolding of events in time as a result of free-will choices, this unfolding of time necessarily occurs also for the cosmic consciousness that is the absolute reality and/or the highest deity and this arises through the free will of both cosmic consciousness and our limited selves.[33]

6 SUBJECTIVITY

What Abhinavagupta's *Svātantryavāda*, Free-Will model adds to these mathematical derivations of free will is his emphasis on subjectivity. This emphasis on subjectivity as necessarily tied to the idea of free will and non-determinism is not found in Gisin's work. Abhinavagupta, on the other hand, includes it it as a central component of his thinking. I suggest that the idea of subjectivity as an element in Abhinavagupta's model intuitively follows from the notion of free will. And even as Gisin is silent on the notion of subjectivity, Brouwer's work on Intuitionist Mathematics, in contrast, turned as well toward the idea of subject, referenced as the "Creating Subject," and pointing to the role of the free choices made by the mathematician as part of the configuration of an Intuitionist Mathematics. Moreover, I suggest that the idea of a foundational subjectivity restructures the way causality operates, and as such, mental causality is written in at the base; it does not need to be derived. Intuitionist Mathematics gestures toward this, offering a mathematical logic premised on free will, the reality of time, and a non-predetermined reality and with this, the element of subjectivity as intrinsic. First, we can briefly look at Abhinavagupta's model to get a sense of his conception of subjectivity and then look at Brouwer's formulation.

Subjectivity is foundational to Abhinavagupta's model of consciousness; it is the marker for consciousness. Subjectivity, the "I," is the basis, which then expands outwardly, taking on a third-person perspective to become the objects we encounter. Abhinavagupta says:

> The Subject's form, which is a unity of awareness contains also an excess, an abundance of awareness. This is deposited into the side of the object that is going to be created. So inwardly the [object] has the attribute of *śakti*, Energy which is none other than the form of consciousness.[34]

To be an object, that is, insentient, is to be in the grip of what we might call an excess of third-person perspective, "this-ness" (*idantā*), in contrast to an innate subjectivity, "I-ness" (*ahantā*), the first-person perspective. Subjectivity as the "I" (*ahantā*) is in essence a form of awareness (*vimarśa*), yet an awareness that intrinsically links to activity (*kriyā*) and with this, is tied to the material world.[35] In Abhinavagupta's system, however, "this-ness" and "I-ness" are modes and as such, not two irreparably separate categories. Rather, subjectivity is embedded at the core of the "this-ness" (*idantā*) that makes up the objects in the world. In other words, subjectivity is the foundation; the third-person perspective is secondary and derived from this.

Even mere objects, a clay jar, at base are simply consciousness as subjectivity. He tells us, "that clay jar—its true nature is the essence of consciousness when it is in a state without mental differentiation; it is complete, like consciousness itself, the body of the whole world."[36] That is, even a mere object has, a deeply retained sense of subjectivity. We see,

> [a]nd in precisely this way, even while accepting the state of being mere object, by properly, in a true way touching and perceiving the object's

> "This-ness" in the highest sense, then out of the objects, those "things" perceived, consciousness alone flows forth. The "thing" in its essential nature is the expanse of Light (*prakāśa*). And this expanse of Light consists of an active awareness (*vimarśa*), the "I" which is not looking towards something other.[37]

Thus at the heart of everything is a foundational subjectivity, premised on a monism. Abhinavagupta draws his monism from his predecessor Somānanda, whose model approximates a pantheism (Nemec 2014). Abhinavagupta adds nuance and the precision of a dual-aspect model to Somānanda's animist nondualist pantheism which sees all things as the expanse of light (*prakāśa*)[38] to formulate his own panentheism as dual-aspect, with both the expanse of light (*prakāśa*) and its inner awareness as subjectivity (*vimarśa*). The light of consciousness (*prakāśa*) congeals to make objects in the world. This is one aspect of his dual-aspect monism. At the heart of this expanse of light—out of which things come to be—at the heart is a subjectivity; this is the second aspect of his dual-aspect monism. This is the "I," an active awareness (*vimarśa*), which is what gives even a clay jar its life, its sentience.

Abhinavagupta explains this dual-aspect monism in cosmological terms, as a cosmic subjectivity, which is the absolute, (*Śiva*), which then unfolds to make our world, yet which nevertheless does not lose its inner consciousness: "pure consciousness lying in the core exists as a state containing both inner and outer. As *Māyā*'s creation spreads out, expands into many branches, the Lord remains still within the core."[39] As *Māyā*'s creation spreads out; we get the complex variety of our world, while still all the while, even in seeming dead matter, clay jars, a pure awareness, consciousness remains at the core. His cosmopsychist model entails a panpsychism—all things are innately conscious—yet they derive from a single unfolding cosmopsychist reality.

As I mentioned, while we do not see the notion of subjectivity in Gisin's work, Brouwer's work, on the other hand, points to a notion of a subject that underlies his model of intuitionist mathematics. He called this the "Creating Subject."[40] As Mark van Atten thoughtfully lays out, Brouwer formulates intuitionist mathematics from a postulation of a subjectivity of the mathematician generating free choices.[41] His idea of the "Creating Subject" focuses on whether the temporal order in which a mathematician establishes mathematical theorems can change the mathematical argument.[42] His arguments suggest that the person, the subject engaging in a mathematical problem within the system of intuitionist mathematics does not exist outside the system, but is, in fact, integral to it. The idea is radical; it eats away at the core of a postulation of objectivity. This turn to an idea of the subject as foundational offers a radically, profoundly different model for understanding the world. As such, it left some of his students perplexed and uncomfortable. His student, Arend Heyting, who later compiled his writings and succeeded him as Mathematics professor at the University of Amsterdam, wrote this regarding Brouwer's idea of the "Creating Subject": "It is true that Brouwer in his lecture introduced an entirely new idea for which the subject is of essential importance. As I said, I feel that it is still questionable whether it is possible and whether it is good to introduce this idea in mathematics" (Niekus 2010: 31). The problem, of course, with bringing in subjectivity is precisely that it upends the notion that science generally, and the mathematical logic that underpins it, offers truth precisely because it proposes a universal, third-person, objective reality divorced from subjective perspectives. When Brouwer brings back in the "Creating Subject" he throws a wrench in the gears of this logic, concluding instead that our external world cannot be calculated and predetermined outside of a subject engaging in mathematical operations.

This entails as well an idea of freedom; as van Atten points out, the "Creating Subject" is essentially free in the choices she makes.[43] This freedom carries with it also again the idea of indeterminism. The Creating Subject cannot know in advance what will come in (van Atten 2018: 1573); in other words, the future is not predetermined, and thus not calculable by an algorithm.[44] It may also be possible to liken Brouwer's understanding of the Creating Subject to a cosmic subjectivity. Iemhoff reads it this way (Iemhoff and Zalta 2020: 2.2), and in this sense, it is akin to Abhinavagupta's formulation of a cosmological cosmopsychism, however, this goes beyond the scope of this paper.[45] In any case, Brouwer offers a way of doing the precise work of mathematics, while retaining the idea of a subjective perspective.

7 CONCLUDING REMARK

I suggest here that it is the shift to a foundational subjectivity that Abhinavagupta makes that underwrites the possibility for a mental causation not reliant on emergentism. The reliance on a subjective perspective goes against our habitual invocation of a third-person perspective as more "logical." However, the mathematical demonstrations of Gisin's and Brouwer's adoption of Intuitionist Mathematics afford a method for inscribing free will, non-determinism and time, and in Brouwer's work particularly, an intrinsic subjective perspective for a philosophy of science. Abhinavagupta's cosmology shares these features with Intuitionist Mathematics, while emphasizing the element of subjectivity as foundational to mental causality. Abhinavagupta's articulation of a subjectively oriented philosophical model can add to our thinking on models of mental causality.

NOTES

1 Abhinavagupta, for instance, asserts causality lies not with material objects, but within a subjective perspective, telling us: "it cannot be proved that the relation of cause and effect is grounded within what lacks sentience. Rather, however, the relation of cause and effect lies only with that which is conscious" (Abhinavagupta 1985a: 3:257) jaḍe pratiṣṭhitaḥ kāryakāraṇabhāvo na upapadyate, api tucidrūpa eva.

2 I discuss this in great detail, with specific attention to understanding Abhinavagupta's relevance for a contemporary philosophy of mind, in Biernacki (forthcoming, 2022), *The Matter of Wonder: Abhinavagupta's Panentheism and New Materialism*, particularly in the fourth chapter. I also discuss the relevance of Abhinavagupta has for contemporary philosophy of mind in Biernacki (2015, 2016).

3 See, for instance, Chalmers (1996). For an examination of different models, see (Goff 2017), and for an extensive overview (Bruntrup and Jaskolla 2016). See as well Strawson (2006) particularly Galen Strawson's work on subjectivity (Strawson 2017).

4 See Sam Coleman's articulate critique of the combination problem and why panpsychism still has a problem with its reliance on emergentism, even as his primary argument focuses on panpsychism more generally (Coleman 2014).

5 See, for instance, Ratie (2014: 171).

6 See, for instance, Shani (2015) and Goff (2019).

7 I outline the details of Abhinavagupta's model of subjectivity at great length in a forthcoming publication.

8 The quote continues, "This happens because of the power (*niyatiśakti*) to determine things as one way or another." In the interest of space I will not go into greater depth here, but propose elsewhere that we may read this as an appeal to theology, and his recognition of the intractability of the decombination problem (forthcoming), even if, as we see below it also entails that even a mere insect can lay claim to this freedom. (Abhinavagupta 1985b: 1:180:) "etadevasvātantryaṃ yadatidurghaṭakāritvam | durghaṭaṃ ca tat niyatiśaktikṛtādeva māyādaśāyāṃ."

9 I discuss at great length Abhinavagupta's model linking subjectivity to freedom (Biernacki, *Matter of Wonder: Abhinavagupta's Panentheism and New Materialism,* Oxford University Press, 2022, forthcoming). Abhinavagupta's cosmopsychism as all-pervasive entails it even in an insect or a clay jar. Abhinavagupta tells us that even for a "clay jar– its true nature is the essence of consciousness" in (Abhinavagupta and Utpaladeva 1986: ad 1.6.3, line 1237) "citsvabhāvo 'sau ghaṭaḥ."

10 Abhinavagupta, *Īśvara Pratyabhijñā Vivṛti Vimarśinī*, ed. Paṇḍit Madhusudan Kaul Śāstrī, vol. 3 (Delhi: Akay Reprints, 1985), 260: "Māyīyānāṃ kīṭāntānāṃ svakāryakaraṇāvasare yat kāryaṃ purā hṛdaye sphurati." See also footnote 55 below, which discusses the distinction Nemec sees between Somānanda and Utpaladeva.

11 This might be understood as akin to Descartes declaration that his thoughts might be true or false, but that his awareness of thinking was indisputable.

12 Bergson wrote of Einstein’s theory of time that it is “a metaphysics grafted upon science, it is not science,” in Canales, Jimena (2015: 6).

13 Gisin (2019: 7, n.10) tells us, “Here I am clearly building on the huge success of quantum information science and on Landauer’s famous claim—‘Information is Physical’, by which he meant that information has always to be encoded in some physical stuff in Gisin (2019: 7, n.10).

14 Gisin points to the disjunct for physicists around the problem of real numbers and infinite information as one that physics mostly rejects in favor of working with the finitude of our real and limited world. Quoting Max Born he says, “My experience is that most physicists reject the faithfulness of such a representation, but admit—often with regret—that they don’t see how one could do otherwise. For example, Max Born, one of the fathers of quantum theory, stressed that ‘Statements like ‘a quantity x has a completely definite value’ (expressed by a real number and represented by a point in the mathematical continuum) seem to me to have no physical meaning,’” in Gisin (2020: 115). And again he tells us, “in practice one never uses real numbers with all their infinitely many digits. Think, for example, of computer simulations that necessarily at each time only consider finite information numbers,” in (Gisin (2020: 116).

15 One of Abhinavagupta’s first twentieth-century interpreters, K. C. Pandey, stressing that Abhinavagupta understood the world as real, labeled Abhinavagupta’s philosophical system as a “Realist Idealism,” in Pandey (1963: 320). Isabelle Ratie argues that the dichotomy of realism/idealism does not work for Abhinavagupta’s Pratyabhijñā and that it does “consider the phenomenal world as perfectly real,” in Ratie (2011: 481, n.4).

16 (Pandey 1963: 320).

17 “Anapahnavanīyatayāvastueva” (Abhinavagupta 1985a: 3:288).

18 Tataḥ kartṛtvaṃ vyatirikte kārye na upapadyate (Abhinavagupta 1985a: 3:316).

19 I unpack the argument he uses in great detail elsewhere (forthcoming).

20 This is in a sense also the point that Karen Barad makes about Niels Bohr’s interpretation of quantum indeterminacy as ontological in contrast to Heisenberg’s uncertainty principle as epistemological, in Barad (2007: 115–16).

21 It is important to note that Abhinavagupta’s formulation here does not entail indeterminacy but at best offers secondary support. I thank Itay Shani for pointing this out.

22 As I note elsewhere (forthcoming), fate, in this way, is never entirely determined, but only partially, by the web of five sheaths (*kañcuka*) that usually overlay our awareness. These include time, a limited sense of knowledge, a limited capacity to do, desire, and fate (*niyati*). By accessing an innate subjectivity, we can be not constrained by sheaths. This means the freedom to change destiny is somewhat accessible, to the extent we can embrace a stance of subjectivity.

23 “Svātantryeṇavā” iti īśvarasṛṣṭyanapekṣayā. Nirmāṇam itisamarthitamitisambandhaḥ (Abhinavagupta 1985a: 3:260).

24 Ipvv vol.3 p. 260: “Nirmātṛpade” iti. Māyīyānāṃ kīṭāntānāṃ svakāryakaraṇāvasare yat kāryaṃ purā hṛdaye sphurati.

25 Abhinavagupta tells us, “Freedom is awareness. This is the reality. There, the impurity occurs by destroying [freedom] through making separate portions,

because this is in fact removing its essential nature. This impurity is a contraction, limiting one's own inherent wealth… For example, if you take a portion of *kajjal*, the black coal from an oil lamp out of its own proper element and place it on some other substance, then there it's just dirt, impurity [making a mess]. But in its own natural habitual state [on the lamp] it is a single unified creation, a form which is one dense mass [of black. Without a counterpoint of comparison] it is completely free from impurity, supremely pure, though that very thing was earlier considered a fault." In Abhinavagupta 1985a: 3:313 "'Svatantro bodha'ityetattattvam. Tatrabhāgāntarahānyā svarūpāpahārāt malamaṇutāṃ saṃkocamādadhānamāṇavaṃ dvidhā. Malaṃhi anyena vijātīyenayogaḥ kajjalaṃśaḥ svadhāyāḥ, tadaṃśo'pi kajjalasyeti ekajātīyamekaghanaṃ rūpaṃ yatnirmalaṃviśuddhaṃ." For the terminology of consciousness as an undifferentiated mass, *cidghana*, see also Bansat-Boudon (2014: 43–4).

26 Emendation to the version from Muktabodha which reads *dhurni* instead of *ghurni*.

27 (Abhinavagupta 1985a: 3:330:) "Svatantra" iti. Svatantrabodho nāma boddharūpo'hamitiyo gīyate, sakaḥ iticintāyāṃ yadādehādinātadrūpaṃ pūryateyogauraḥ, so'haṃ; yadṣuddhitaḥ, so'hamityādi; tadādehādirevapradhānībhavati saṃvitsvarūpaṃ nyakkṛtaṃ jātamitibhedyameva pradhānībhavati, na vedakarūpam, yadātudehādeḥ kiṃtattvamiti cintopakramaṃ prakāśaghanamevasaṃvidrūpamiti, tadābodhassvarūpīkṛtaṃ tadrasānuviddham eva śūnyādidehāntamavabhātīti abhyāsāt tasyasaṃviddharmāḥ śaktiviśeṣāḥ samyagāviśanto vibhūtīrutthāpayanti, anabhyāse'pitutatkṣaṇāveśa eva ānando dbhavakampanidrāvyāptirūpaghūrṇyāvirbhāvanakrameṇa jīvanmuktatālābhaḥ (Abhinavagupta 1985a: 3:330).

28 Somadeva's Kathāsaritsāgara offers a nice illustration of these powers of a yogi, in (Somadeva 1915, *passim*).

29 "Yadātutasyaciddharmavibhavāmodajṛmbhayā |vicitraracanānānākāryasṛṣṭipravartane|| bhavatyunmukhatācittāsecchāyāḥ prathamātuṭiḥ | (2/8) iti. (Abhinavagupta 1985a: 3:263) Paramaśive'pi bhagavati aunmukhyarūpāyānandaśakterīcchāśakty apekṣayā prathamatvaṃ pūrvabhāvitvaṃ, tuṭiritikālāṃśarūpatvaṃ tadupapadyate; anyathākramābhāve kathaṃ pūrvāparībhāvavyavahāraḥ kintu, upadeśyāpekṣayā antargatamāyīyapramātrapekṣayā cayukto'sāviti.

30 Nanu atrasūtrekramavanto māyāpramātāraḥ sṛṣṭāḥ, natu śivatattvam (Abhinavagupta 1985a: 3:263). Āha "pramātṛsāmānyāt" iti. Tatrāpihi upapāditākramasthitiḥ. Ata eva uktaṃ "tad evam" iti.

31 In a gesture that seems to accommodate both views, as Gisin does with his nuanced notion of two types of time, Abhinavagupta tells us it depends on our perspective. He gives us alternative readings of Utpaladeva's text on which he is commenting here: "Alternatively, he says, 'Or perhaps' these souls, these *Māyā*-bound souls do not at all [have sequence either] here viewing them from the perspective of the ultimate reality" (Abhinavagupta 1985a: 3:263) "Tevānakeciditivāstavaṃ pakṣamāha 'athavā' iti." If the divine itself is free of time, then we might suggest that viewed from this perspective ordinary beings are as well. This last point does indeed offer a counter-argument. I suggest that this is characteristic of Abhinavagupta's style, to give us a rich presentation of possibilities that do not all agree. I thank Susanne Beiweis for pointing this out.

32 As Gisin points out, tied to these ideas of free will and time unfolding is also a logical refutation of the Law of the Excluded Middle. He draws from Brouwer here as well (Brouwer 1981: 5–8). Gisin tells us that this Law depends upon the assumptions of infinite information tied to real numbers and that if we adopt a different mathematical model, that of intuitionist mathematics, the Law of the Excluded Middle no longer holds, in Gisin (2020: 116). The example he uses has to do with the incorporation of time, for instance, the question of whether it will be raining in a year or not is not amenable to the Law of the Excluded Middle. Given the use of this concept as a way of criticizing Indian philosophy, Gisin's observation of the assumptions entailed in the Law of the Excluded Middle is interesting. However, Abhinavagupta's adoption of structural similarities to Gisin's intuitionist mathematics is not found in other Indian philosophies, such as Advaita Vedānta.

33 George Dreyfus articulates this idea I think as well from a Buddhist perspective, when he suggests that we leave phenomenology when we posit a transcendent self, in Dreyfus (2013: 135).

34 Abhinavagupta, *Ipvv*, 3:258: Bodhaikarūpasya pramātur yad adhikamabodharūpaṃ, tat nirmātavyapakṣe nikṣipyate iti tasya yat nijam śaktirūpaṃ dharmaḥ.

35 Elsewhere, I translate *vimarśa* as "active awareness" and discuss the relationship between *vimarśa* as an active awareness and *prakāśa* as the elment of light which gives rise to the appearance of our material world in great detail (Biernacki 2022). Given the complexity of the argument and the constraints of space I will not go into this argument here. I also there discuss the relation these terms bear to the terms *jñāna*, knowledge, and *kriyā*, action.

36 "Tadavikalpadaśāyāṃ citsvabhāvo 'sau ghaṭaḥ cidvadeva viśvaśarīraḥ pūrṇaḥ" (Abhinavagupta and Utpaladeva 1986: ad 1.6.3, line 1237). See here also Catherine Prueitt's extended discussion of this passage relating this to the concept of apoha, differentiation as delimitation in Prueitt (2017).

37 Tathā hi vedyadaśām aṅgīkṛtavatām api ata eva ucitena idantāparāmarśena parāmṛśyānāṃ bodha eva prakāśātmā sāraṃ vastu, prakāśaś ca ananyonmukhāhaṃ vimarśamayaḥ (Abhinavagupta 1985a: 3:274). Here again Abhinava indulges in a play on words where parāmarśa means both touching the "other" and touching the highest. And again he playfully contrasts this with the word "*vimarśa*" the supreme power of active awareness, the "touching" of the Lord.

38 Abhinavagupta derives this dual-aspect monism from Utpaladeva's writings; clearly present in Utpaladeva's work, it is less clear in Somānanda's writings.

39 Abhyantaracinmātrāpekṣayā antarbahirbhāvasya sadbhāvaḥ, māyīye tu sarge >bhy antareśvaravṛddhyādyapekṣayā bahuśākho >sau (Abhinavagupta 1985a: 3:262).

40 See particularly Mark Van Atten's helpful analysis of Brouwer's "creating subject" in van Atten (2018). See also Niekus (2010). The mathematics involved in both of these articles is beyond my skill set, however, both of these authors, van Atten in particular, do a good job of situating these mathematical proofs within a contextual argument to make them accessible for broader philosophical interest.

41 As van Atten (2018: 1594) points out here, quoting Troelstra, intutionist mathematics entails the "subjectivistic viewpoint of intuitionism."

42 As van Atten points out, Brouwer's intuitionist mathematics recognizes the foundational priority of the subject. Van Atten tells us, "the Creating Subject cannot simply accept a proof from outside, and must go through the steps to reconstruct the evidence. In the end, then, its own ability to make A evident is what really matters," in (van Atten 2018: 1595).

43 Van Atten makes this point within a mathematical formulation, and it may be of interest for mathematically inclined readers to examine the mathematical proof he lays out in detail; for our purposes here, I focus on the philosophical idea of subjecitivy underlying the model. I reproduce here van Atten's mathematical formulation, symbols intact, for interested readers, but the mathematical proof is beyond my ken: "Then it can be argued from the essential freedom of the Creating Subject, that ∀n°˛°˛∃k(a*n (k) = 1) (112) because the Creating Subject is free to work towards constructing its nth lawlike sequence, but that ∀n ∃k(a*n (k) = 1) (113) is not true, because the same freedom allows the Creating Subject, for given n, not to begin constructing its nth lawlike sequence by whatever stage k the lawlike sequence a*n indicates," in (van Atten 2018: 1609). Also van Atten reiterates the importance of freedom, the free choices of the "Creating Subject" saying, "Troelstra is clear that Creating Subject sequences ('empirical sequences') do depend on free choices of the Creating Subject, and that they are not lawlike in a sense that would include their being independent of the Creating Subject's free choices," again in (van Atten 2018: 1609).

44 This brings in again the idea of time, as a separate argument from what we saw in Gisin's argument for including time in a mathematical schema. As van Atten points out, we find this in Brouwer: "To see why the time parameter in this sense is mathematically relevant for the Creating Subject, recall Brouwer's statement in his dissertation that mathematics is first of all an act." I am reminded here of Abhinavagupta's statement that we saw earlier, linking "Doership" to the reality of the world. That is, the world is real because it is linked to the willful actions of a subject. In this case Abhinavagupta shares with Brouwer a recognition of the fundamental importance of instantiated, material activity as an element of a first-person perspective.

45 Van Atten addresses this element of reading Brouwer's Creating Subject as a singular subject, as well, and quotes Brouwer to argue, in effect, against this reading, suggesting rather that the Creating Subject be understood schematically, with a possible instantiation in a variety of subjects, in (van Atten 2018: 1595). In both cases, I think what is key is the shift in the model from a third-person perspective to subjectivity as the grounding perspective.

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